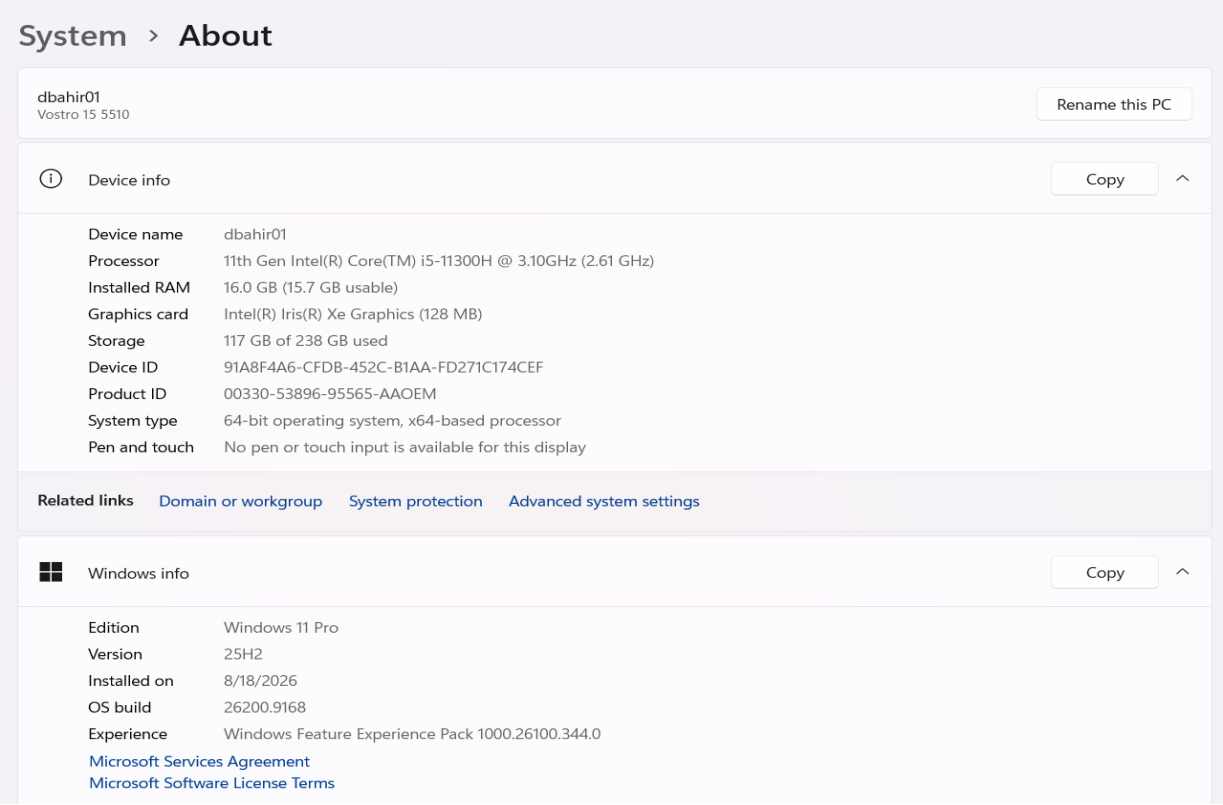
Basic computer hardware

Tasks :-

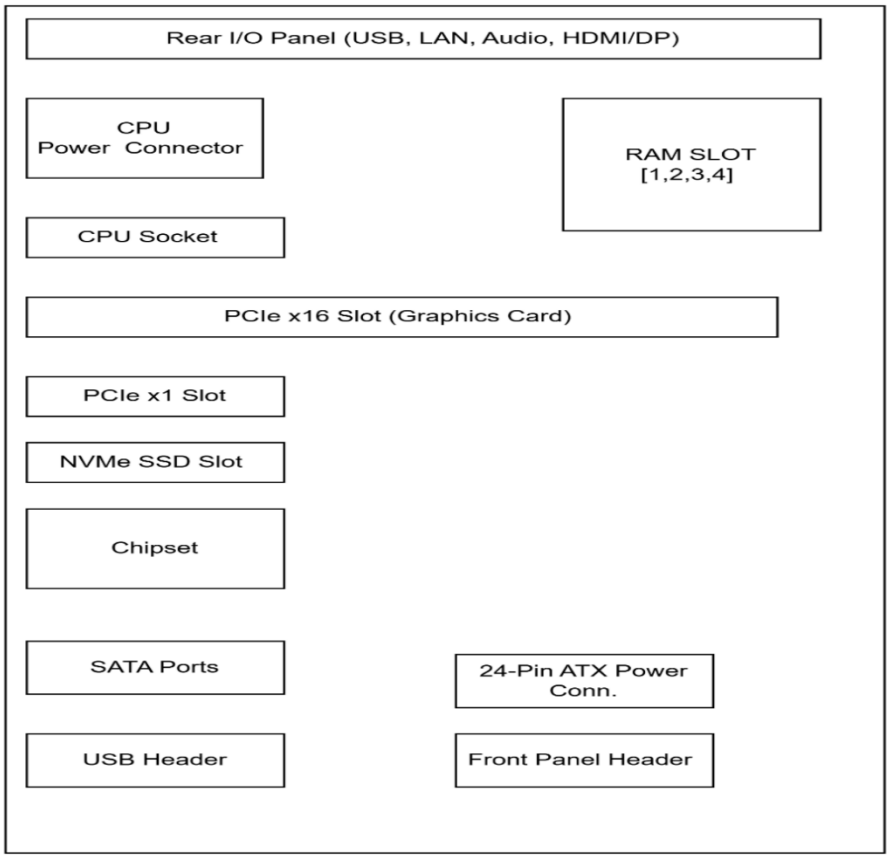
1.Open your computer or laptop and take a clear photo or screenshot of the System Information or About page that shows your processor (CPU), RAM, and storage details.

Ans :-



2.Draw and label a diagram (on paper or digitally) showing the main hardware components inside a computer cabinet, including CPU, RAM, motherboard, hard drive/SSD, power supply, and at least two ports (USB, HDMI, etc.).

Ans :-



**3.List the hardware components you would check or upgrade to make your laptop run faster for gaming or video editing, and explain in one line why each upgrade helps.

Hint: Think about RAM, SSD, GPU, and cooling.**

Ans :-

- **RAM (Memory)** – Upgrade to at least 16GB to prevent stuttering and lag when handling large textures, timelines, and multitasking.
- **SSD (Storage)** – Swap to a fast NVMe M.2 SSD to slash load times, boot speeds, and render-file read/write delays.
- **GPU (Graphics Card)** – If your laptop has a replaceable MXM GPU, upgrade it (or buy a new laptop with a better one) to dramatically accelerate 3D rendering, encoding, and frame rates.
- **Cooling System (Fans/Thermal Paste)** – Clean fans and reapply high-quality thermal paste to avoid thermal throttling, which slows down both CPU and GPU under heavy loads.
- **CPU (Processor)** – Only if socketed (rare), upgrade to a higher-tier model to speed up physics, effects, and codec decoding/encoding.
- **Power Adapter** – Check that your AC adapter provides enough wattage; underpowering can cap performance during sustained gaming or rendering.

4.Research the difference between SSD and HDD storage, then write a short comparison (3 points) explaining which one is better for a student who wants fast boot times and why.

Ans :-

Feature	SSD	HDD	Why SSD is Better for Boot Times
Speed & Boot Time	5 to 15 seconds	30 to 90 seconds or more	SSDs use flash memory with no moving parts, allowing them to read data almost instantly, which drastically cuts down the time it takes for your operating system to load .
How It Works	Uses flash memory chips for data storage	Uses spinning magnetic platters and a moving read/write head	Because an SSD doesn't have to rely on mechanical parts to find data, it can access files nearly instantly, whereas an HDD must wait for the platter to spin and the head to move into position .
Durability	No moving parts, more resistant to drops and shocks	Moving parts are fragile and prone to failure from physical shock	For a student carrying a laptop between classes, an SSD is less likely to be damaged by accidental bumps, which also protects your data .